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POLICY BRIEF

MAIZE VALUE CHAIN IN KENYA:
IMPLICATIONS ON MAIZE MILLING COSTS

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1. Introduction

Maize is one of the strategic agricultural commodities which has been prioritized in the Agricultural Sector Transformation and Growth Strategy (ASTGS 2018-2028) of the Ministry of Agriculture, Livestock and Fisheries (MoALF, 2019). It plays a central role in contributing to food security, trade and livelihoods of millions of people in Kenya. Maize is the cheapest source of calories among the Kenyan households. It accounts for 24 per cent of all calories consumed and 50 per cent of staple food calories consumed (KNBS, 2020). Maize accounts for only about 2.3 per cent of the total value of agricultural production, other commodities (e.g. cattle, milk and tea) are more important by value.

2. Domestic demand for maize

Maize continues to be the most important staple food in Kenya as demand is growing, driven by population growth. The demand for maize currently stands at 52 million bags, and it has been projected to reach 60 million bags by 2025. **Table 1** shows Kenya's maize supply and disposition for the period 2010-2019. The highest self-sufficiency ratio the country has ever attained is 97% which was witnessed back in 2013. The estimated maize consumption is about 67.3kg/capita/year in Kenya (KNBS, 2020).

Table 1: Kenya Maize Supply and Disposition, 2010-2019 (1000 tonnes)

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Production	3,222	3,377	3,600	3,501	3,507	3,824	3,406	3,186	4,014	3,897
Imports	349	314	325	94	473	489	164	1385	564	258
Exports	1	15	1	2	16	15	15	18	8	4
Consumption	3384	3,609	3,789	3,996	4,032	4,059	4,203	4,320	4,479*	4,479*
Available Supply	3,157	3,516	4,469	4,138	3,774	3,805	3,349	4,148	4,147	3,743
Self-sufficiency ratio	90.25%	91.87%	91.74%	97.44%	88.47%	88.97%	95.81%	69.98%	87.83%	93.88%

· *Calculated estimates*

Source: KNBS, 2020

3. Maize value chain in Kenya: From Inputs to distribution

Input supply: Adoption of improved seed varieties is approximately 80% from a national variety list of 363 maize varieties. The market continues to be dominated by Kenya Seed Company, a parastatal that holds a market share of 70–80% (Christinck et al., 2018). The volume of certified seed produced has increased from 27,078 metric tonnes (MT) in 2008 to 54,555 MT in 2017. Despite this growth, the country still imported an estimated 7,000 MT of seed in 2017 from Zambia (KEPHIS, 2019).

Maize farmers in Kenya use an average of 30kg/ha fertilizer, which is far below the 50kg/ha recommended under the Abuja Declaration of 2006. The country's fertilizer market is fully liberalized with the bulk of fertilizers imported and distributed by the private sector.

Production: Maize is grown on an estimated 1.4 million hectares in Kenya by large-scale farmers (25 percent) and smallholders (75 percent). It is both a commercial and subsistence crop with an average annual output estimated at approximately 3 million metric tons, giving a national mean yield of 2 metric tons per hectare (KNBS, 2020). Parts of the Rift Valley Province in western Kenya, particularly the counties of Trans Nzoia and Uasin Gishu, produce a large maize surplus, primarily on medium and large farms. Key production challenges include use of right seeds and fertilizers, better crop husbandry, aflatoxin, lack of extension services to farmers, low soil fertility, pests and diseases.

Aggregation and trade: Value chain integration is not widespread in Kenya; a network of small and medium sized traders and wholesalers buy maize from farmers and sell it to processors. This node has control of the value chain vested in knowledge of prices and availability, aggregation and buyers.

Post-harvest handling, storage and management: The major actors include producer cooperatives, small and medium-sized traders and processors that have vertically integrated into this stage of the chain. The main incentive of this stage of the value chain is to achieve economies of scale so that farmers can satisfy large procurements by processors and thus, get value for their produce.

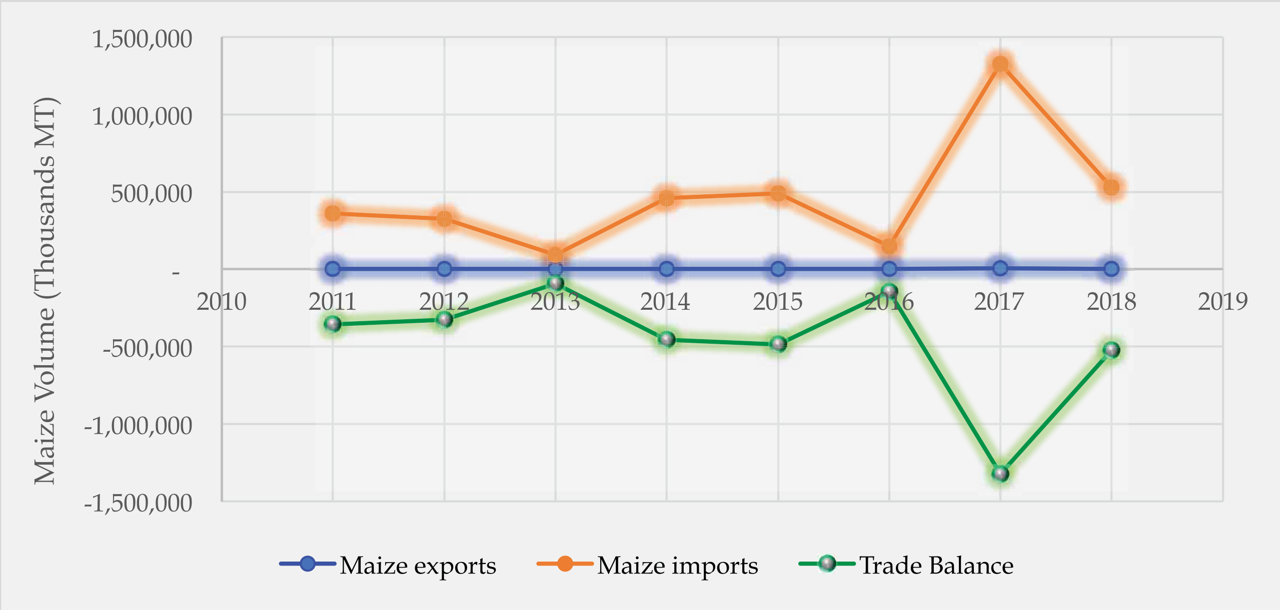
Processing: Dry mill technology is dominant in Kenya; Hammer millers (Posho millers) offer milling services to the farmers for producing un-sifted maize meal. Their milling capacity of is between 6 and 110 metric ton per month, but utilization is estimated at about 15% (KAVES-USAID 2014). With a capacity of 10 to 200 metric tons per month, small-scale millers produce sifted maize meals and maize by-products such as the composite maize meal. Medium and large-scale millers account for about 93% of the annual total installed milling capacity, estimated at 1.4 million metric tons. The four main large-scale millers (Mombasa, Unga, Pembe and Eldoret) account for 70% of total milling capacity. The millers produce sifted maize meals in several packages (1 kg, 2 kg, 5 kg, 10 kg and 50 kg). Apart from sifted maize meal, these millers are also involved in animal feed production. Most of the millers are operating below their installed capacity due to short supply of maize occasioned by production challenges and post-harvest losses.

4. Maize Marketing and trade

Maize marketed sales in Kenya are highly concentrated. In the country, about 57% of small-scale farmers are net-buyers of maize, while 11% are purely subsistence producers. Even though small-scale farmers produce 75% of the maize, that is consumed within the households, and they only sell approximately 20% of their production, (Jayne et al., 2010; Kirimi et al., 2011; Abate et al., 2015).

Maize in Kenya has traditionally been a domestic market product with very minimal exports. Kenya is a net importer of maize due to higher consumption relative to domestic production. The average maize production over the past 15 years has been estimated at 3.2 million metric tons, while consumption was 3.5 million metric tons (ERA, 2015; OECD-FAO, 2016). As illustrated by **Figure 1**, maize imports have increased by 47 % between 2011 and 2018, with peak imports in 2017 during the regional maize shortage that resulted in an import surge from international markets. The maize trade deficit increased to over 526,000MT in 2018 from 358,000MT in 2011. From the maize exports and imports trend, it is evident that Kenya operates under the import price parity trade regime.

Figure 1: Maize imports, exports and trade balance, 2011-2018



Source: KNBS Economic Survey 2016 and 2018

Cross-border trade flows are aligned to the respective crop calendars for Kenya and its main trading partners (Uganda and Tanzania). Peak imports are usually during the second and third quarters of the year, which coincide with the lean period domestically and harvests from trading partners. Imports from Uganda typically peak in the third quarter while those from Tanzania peak during the second quarter. Conversely, import volumes are much lower during the first and fourth quarters due to relatively high domestic availability of maize from the harvest season which takes place between October and December. **Figure 2**, illustrates these trends based on data on informal cross-border trade.

Figure 2: Five -year average monthly imports from Uganda and Tanzania, 2014-2018

Source: Computed from EAGC RATIN and FEWSNET Data, 2019

The regional crop (harvest) calendar as illustrated by **Table 2**, exerts a significant influence on trade patterns, particularly where harvests in surplus maize producing countries coincides with lean periods in deficit countries, thus creating a natural opportunity for trade from the surplus countries to Kenya.

Table 2: Regional calendar for main maize harvests

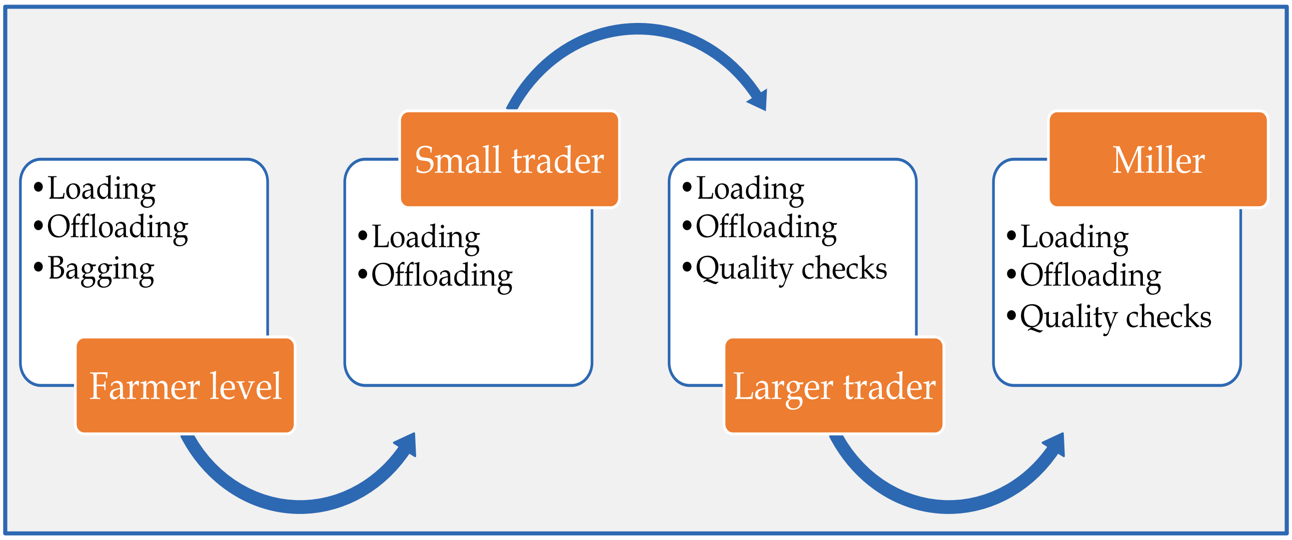
	J	F	M	A	M	J	J	A	S	O	N	D
Kenya												
Tanzania												
Uganda												
Burundi												
Rwanda												
S/Sudan												
Ethiopia												

Source: EAGC, 2019

5. Multiple levels of maize grain handling: Cost implications to millers

At each level of the maize value chain, a handling cost is charged on each bag of maize **(Figure 3)**. Loading and offloading costs are standard across all levels; quality checks are more common at trader and miller level. Cost of bagging are also prominent at the farmer and/or trader level. Stakeholder interviews revealed that these handling costs could add up to KES 200 to the cost of each bag before milling, equivalent to 8% of the price of a bag of maize in Eldoret[1].

Figure 3: Multiple levels of grain handling from farm to miller



Source: EAGC, 2019

1 *Computed from EAGC RATIN database using Eldoret wholesale maize price on 19th December 2019.*

Table 3: Cost build-up for maize flour production

Item	Per Kg	Distribution
Cost of maize	36.72	75%
Handling	3.32	7%
Logistics	1.31	3%
Energy	1.26	2%
packaging	3.74	8%
Labour	1.65	3%
Other	1.11	2%
Total	49.47	

Source: Survey findings, 2019

Key cost drivers for maize milling

The cost of maize grain accounts for approximately 75% of maize milling costs. Other important cost components are packaging costs (about 8% of milling costs) and handling costs which include cleaning, drying, storage and quality control (7 % of milling costs).

On average, millers incur KES 50 to produce a kilogramme of maize flour, exclusive of other overhead charges (such as regulatory compliance, taxes, etc.). **Table 3** summarises the cost build-up for maize flour production and their distribution percentage.

6. Impact of high maize costs on maize milling and food prices

Table 4: Cost of milling maize flour, assuming a 50% reduction in maize costs

Item	Per Kg	Distribution
Cost of maize	36.72	18.36
Handling	3.32	3.32
Logistics	1.31	1.31
Energy	1.26	1.26
packaging	3.74	3.74
Labour	1.65	1.65
Other	1.11	1.11
Total production	49.47	30. 75
millers Margin	10.00	10.00
Ex-factory price	59.47	40.75
Retail price	69.47	50.75

Source: Computations from survey findings, 2019

Considering that the cost of maize is the most significant cost component in maize flour, reducing the cost of maize for millers bears the most fruit in terms of lowering milling costs and delivering more affordable flour to consumers. The cost of production of maize is approximately half that in Uganda. Assuming Kenyan maize was equally cheap to produce as Ugandan maize, it this effectively means that the cost of maize delivered to millers would also reduce by approximately 50 %, which will allow maize flour prices to retail at about KES100 per 2kg packet, other factors remaining constant (see **Table 4**). This means that significant efforts need to be directed, both by public and private sectors, to improve the productivity of Kenyan farmers and reduce their transaction costs so that they can deliver cheaper maize to millers without compromising their incomes.

7. Impact of post-harvest losses, handling and storage challenges on maize millers

Post-harvest losses, handling and storage challenges at farm level affect millers in three main ways:

i. **Reduce the volume available for milling:** The 2019 season is a case in point; the rains undermined the pace of harvest. Due to this, the maize market remained thin even during the harvest period. This meant that several millers were operating below their capacity. At the time of this survey, most millers indicated that they were operating at 36 % capacity.

ii. **Increases the price of maize for millers:** After (i) above, maize prices remained high throughout the harvest period. At the time of this study, millers in Nairobi and Thika were paying KES 3,500 per bag for maize deliveries, while the prevailing price in Eldoret was approximately KES 2,700 per bag. By comparison, in the same period in 2018, Nairobi/Thika and Eldoret prices were approximately KES 2,202 and KES 1,707 per bag respectively[2].

iii. **Comprises the quality of maize:** Taking deliveries of wet maize increases the risk of aflatoxin contamination and prevalence of rotten and diseased grains. The quality of maize is a leading concern amongst millers, followed by the cost of quality control (laboratory costs).

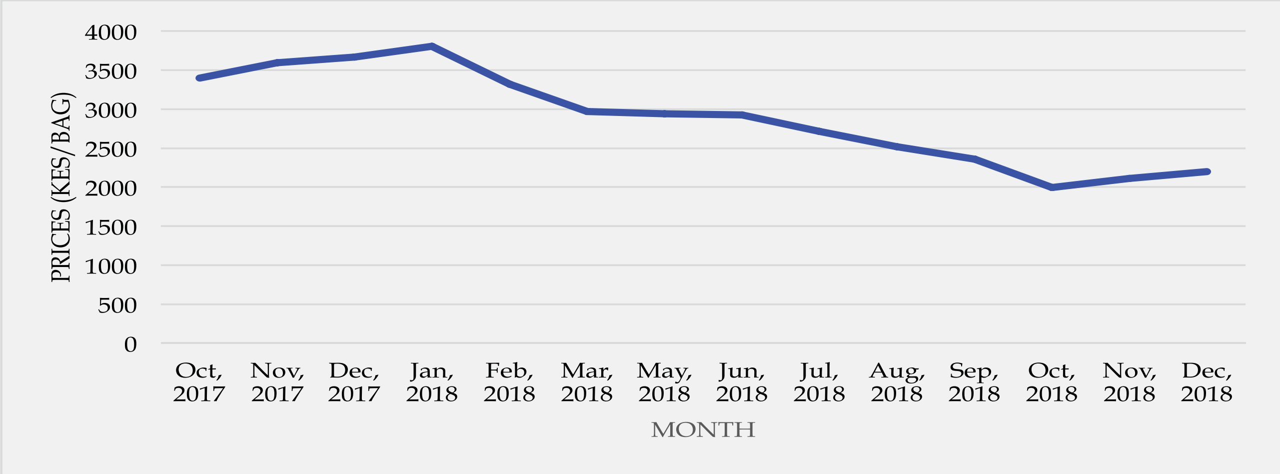
8. Key policy challenges affecting maize millers

The cross-border trade environment: The results of this study found that the time taken in moving goods across borders has improved over the past five years. However, the cost of doing business across borders has increased during the same period. For example, exports of some food products derived from grains to the other EAC Partner States attract VAT and other local taxes, despite those taxes not being charged in Kenya. Consequently, the miller cannot pass on this cost to suppliers or customers and thus must absorb them in current business margins. This shows that investments to facilitate trade, such as One-Stop Border Posts have succeeded to reduce the time taken to clear goods across borders but have not reduced the cost of conducting such business.

Government participation in domestic markets: The Government, through the Strategic Food Reserve Trust Fund (SFRTF) and NCPB, plays an integral but somewhat controversial role in local maize markets. The most unfavourable aspects of government participation in markets are price-setting practices. The price-setting does not consider the price discovery process that reflects the cost of production; prevailing market prices (national and regional) and an appropriate percentage mark-up for farmer profits. During harvest, maize prices become artificially inflated, particularly at the time of harvest because NCPB/SFRTF become the buyer of first resort for farmers. For instance, during the harvest season in 2017, the Government announced that it would purchase maize from farmers at KES 3,200 per 90kg bag through SFRTF and its agent, NCPB. At the same time, the average maize price in Eldoret and Nakuru was between KES 2,600 and 2,800 per 90kg bag. Subsequently, prices in Nairobi increased by 12 % between October 2017 and January 2018. With limited capacity by NCPB to absorb the surplus and millers opting for cheaper Tanzanian and Ugandan maize, the prices declined significantly for the remainder of the year; prices in December 2018 were 42 % below January 2018 prices **(Figure 5)**.

2 *Computed from EAGC RATIN database.*

Figure 5: Nairobi Maize prices, 2017/18



Source: EAGC RATIN, 2019

Inferior quality of maize distributed from NCPB depots to millers: Millers point out that the quality of maize distributed by NCPB is usually of inferior quality and occasionally unfit for human consumption. This is backed by a report by KEBS in 2018 which revealed that 63.3 % of maize in NCPB silos, worth over KES 7 billion at the time, was unfit for human consumption[3]. NCPB also acknowledges challenges with the quality of maize in its depots, noting that the Board does not have adequate facilities to monitor grain quality as it has only one laboratory to cater for its facilities scattered in 46 of the 47 counties in the country (EAGC, 2018; NCPB, 2018).

Lengthy bureaucracies at NCPB: Millers pointed out that the process of collecting maize stocks from NCPB depots is fraught with bureaucracies through which corrupt practices manifest. Specifically, millers pointed out the following bottlenecks:

a. Long queues at NCPB depots to collect maize whereby millers wait up to 3 days. During that time, they must cover the costs of accommodation and meals for their staff assigned to buy the maize.

b. Some NCPB staff allegedly demand bribes to allow the speedy collection of maize from the depot, even if the miller has fulfilled all the requirements.

c. Similar to (b), some of the bags from NCPB stores were found to be underweight. In one case, a miller in Nakuru who bought 200, 50 kg bags of maize from NCPB later found that they weighed less by between 5kg to 10kg. Assuming there was a 5kg loss in each bag, this amounted to a 1MT of maize lost.

9. The overall impact of the business environment on milling costs

Table 5: Ranking of key business environment factors for maize milling

Issue	Rank (1=most importand)
Compliance with regulations, fees, levies and taxes	1
Government participation in markets	2
Compliance with Standards	3
Cross-border trade costs	4

Source: EAGC, 2019

In this study, millers were unanimous that the costs of milling have increased over the past five years. The most important factors behind the increase in costs are (in order of impact) compliance with regulations (fees, taxes and permits), government participation in markets (modalities for purchase and sale of maize through strategic food reserve), compliance with quality standards and cross-border trading costs (Table 5). Other areas which affect milling costs but to a lesser extent are costs of energy and trade finance.

3 *See https://www.nation.co.ke/news/State-orders-fresh-audit-on-bad-maize-stored/1056-4840204-124w2riz/index.html*

10. Recommendations for reducing maize milling costs

1. **Improve the market system for maize trade through structured trading systems**

   Adoption of structured trading systems for grain trading to improve efficiency in maize trade through the Warehouse Receipt System (WRS) that offers millers cost reduction, quality and quantity guarantee, reduced multiple handling of grains and price risk management and procurement planning**.**

2. **Tax proposals to reduce maize milling costs by: -**
   - Allowing duty-free yellow maize specifically for animal feeds under duty remission scheme
   - Removing taxes on post-harvest handling equipment, products and services
   - Abolishing multiple cesses and similar charges of maize consignments

3. **Public-Private Partnerships to reduce post-harvest losses and improve grain handling and marketing in the maize value chain through: -**
   - Restructuring NCPB into a public-private commercial entity
   - Engage the private sector for storage for food stocks on behalf of Strategic Food Reserves (SFR)
   - Leasing NCPB facilities to private sector operators
   - Purchase of maize for the SFR through trade platforms and commodity exchanges

4. **Strengthen data collection and information sharing to inform policy and business decisions**

5. **Strengthen implementation and compliance mechanisms of quality standard for improved food safety along maize value chain**

11. Implementation Matrix

	Policy	Action	Responsible Institution	Short Term	Medium Term	Long Term
1	**Improve the market system for maize trade through structured trading systems**	Adoption of structured trading systems for grain trading	Government of Kenya- State Department of Trade Nairobi Stock Exchange- Kenya Multi Commodity Exchange (KOMEX)		X	
2	**Tax proposals to reduce maize milling costs**	Allowing duty-free yellow maize specifically for animal feeds under duty remission scheme	Government of Kenya - KRA, MoALF&C	X		
		Removing taxes on post-harvest handling equipment, products and services	Government of Kenya - KRA, MoALF&C	X		
		Abolishing multiple cesses and similar charges of maize consignments	Government of Kenya and County Governments		X	
3	**Public-Private Partnerships to reduce post-harvest losses and improve grain handling and marketing in the maize value**	Restructuring NCPB into a public-private commercial entity	Government of Kenya		X	
		Engage the private sector for storage of food stocks on behalf of SFR	Government of Kenya		X	
		Leasing NCPB facilities to private sector operators	Government of Kenya		X	
		Purchase of maize for the SFR through trade platforms and commodity exchanges	Government of Kenya, Industry Players		X	

	Policy	Action	Responsible Institution	Short Term	Medium Term	Long Term
4	**Strengthen data collection and information sharing to inform policy and business decisions**	Improve coordination of data collection, and data sharing	Government of Kenya-KNBS, County Governments, Research Organisation, Universities, EACG, Industry Players		X	X
5	**Strengthen implementation and compliance mechanisms of quality standard for improved food safety along maize value chain**	Strengthen the provision of extension services at National and County levels Promote self-regulation to enforce adherence to food safety standards	Government of Kenya-KNBS, County Governments, Research Organisations, Universities, EACG, Industry Players	X	X	

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